

Seminar 22/10

October 2024

1 Problem 1

Consider the following data about some project:

- \$ 300 000 on a feasibility study (already paid for).
- Life-time of the project: 5 years (year 1,2,..., 5).
- Sales forecast: 200 000 units per year.
- Wholesale price: \$ 350.
- R&D cost: \$ 25 million in year 0.
- Unit production cost: \$ 250.
- Upfront investment in new equipment (year 0): \$ 6.3 million. Life-time: 5 years (use straight line depreciation).
- Yearly marketing cost, years 1 to 5: \$ 4 million.
- AR is 25% of annual sales, AP is 25% of annual COGS.
- At the end of the 5th year, the project is sold at book value.
- $r = 12\%$.
- Tax rate: 40%.

1. Calculate the incremental FCF of the project.
2. Calculate the project's NPV.

Solutions:

1. We will build the incremental earnings statement, and follow it with the incremental cash flow statement. All values are in thousand dollars.

	0	1	2	3	4	5
Revenue		70 000	70 000	70 000	70 000	70 000
COGS		(50 000)	(50 000)	(50 000)	(50 000)	(50 000)
Gross Profit		20 000	20 000	20 000	20 000	20 000
SG&A		(4000)	(4000)	(4000)	(4000)	(4000)
R&D	(25 000)					
Depreciation		(1260)	(1260)	(1260)	(1260)	(1260)
EBIT	(25 000)	14 740	14 740	14 740	14 740	14 740
Tax	10 000	(5896)	(5896)	(5896)	(5896)	(5896)
NOPLAT	(15 000)	8844	8844	8844	8844	8844
+ Depreciation		1260	1260	1260	1260	1260
- CAPEX	(6300)					
- Change in NWC		(5000)	0	0	0	5000
FCF	(21 300)	5 104	10 104	10 104	10 104	15 104
PV (FCFF)	(21 300)	4 557	8 055	7 192	6 421	8 570
NPV	13 496					

2 Problem 8-24 Seminar 6

Bauer Industries is an automobile manufacturer. Management is currently evaluating a proposal to build a plant that will manufacture lightweight trucks. Bauer plans to use a cost of capital of 12% to evaluate this project. Based on extensive research, it has prepared the following incremental free cash flow projections (in millions of dollars):

	Year 0	Years 1–9	Year 10
Revenues		100.0	100.0
– Manufacturing expenses (other than depreciation)		–35.0	–35.0
– Marketing expenses		–10.0	–10.0
– Depreciation		–15.0	–15.0
= EBIT		40.0	40.0
– Taxes (35%)		–14.0	–14.0
= Unlevered net income		26.0	26.0
+ Depreciation		+15.0	+15.0
– Increases in net working capital		–5.0	–5.0
– Capital expenditures	–150.0		
+ Continuation value			+12.0
= Free cash flow	–150.0	36.0	48.0

- a. For this base-case scenario, what is the NPV of the plant to manufacture lightweight trucks?

- b. Based on input from the marketing department, Bauer is uncertain about its revenue forecast. In particular, management would like to examine the sensitivity of the NPV to the revenue assumptions. What is the NPV of this project if revenues are 10% higher than forecast?
- c. Rather than assuming that cash flows for this project are constant, management would like to explore the sensitivity of its analysis to possible growth in revenues and operating expenses. Specifically, management would like to assume that revenues, manufacturing expenses, and marketing expenses are as given in the table for year 1 and grow by 2% per year every year starting in year 2. Management also plans to assume that the initial capital expenditures (and therefore depreciation), additions to working capital, and continuation value remain as initially specified in the table. What is the NPV of this project under these alternative assumptions?
- d. To examine the sensitivity of this project to the discount rate, management would like to compute the NPV for different discount rates. Create a graph, with the discount rate on the x-axis and the NPV on the y-axis, for discount rates ranging from 5% to 30%. For what ranges of discount rates does the project have a positive NPV?

Solutions.

a.

$$\begin{aligned}
 NPV &= \sum_{t=0}^{10} FCF_t \\
 &= -150 + 36 \times \sum_{t=1}^9 \frac{1}{(1.12)^t} + \frac{48}{(1.12)^{10}} \\
 &= -150 + 36 \left(\frac{1 - \frac{1}{(1.12)^{10}}}{1 - \frac{1}{1.12}} - 1 \right) + \frac{48}{(1.12)^{10}} \\
 &= 57.27
 \end{aligned}$$

- b. If the Revenues are 10% higher, then they equal 110. Hence, the EBIT will be 50, and the unlevered income will be $50 \times 0.65 = 32.5$. In years 1 to 9, the FCF will be 42.5; in year 10, it will be 54.5. Hence, the NPV will be:

$$\begin{aligned}
 NPV &= \sum_{t=0}^{10} FCF_t \\
 &= -150 + 42.5 \times \sum_{t=1}^9 \frac{1}{(1.12)^t} + \frac{54.5}{(1.12)^{10}} \\
 &= -150 + 42.5 \left(\frac{1 - \frac{1}{(1.12)^{10}}}{1 - \frac{1}{1.12}} - 1 \right) + \frac{54.5}{(1.12)^{10}} \\
 &= 94.
 \end{aligned}$$

- c. For any year $t \in \{1, \dots, 10\}$, an expression for the Unlevered net Income is:

$$(55 \times (1.02)^{t-1} - 15) \times 0.65 = 35.75 \times (1.02)^{t-1} - 9.75$$

In year 0, the FCF is the same (-150). In any year $t \in \{1, \dots, 9\}$, an expression for the FCF is:

$$FCF_t = 35.75 \times (1.02)^{t-1} - 9.75 + 10 = 35.75 \times (1.02)^{t-1} + 0.75$$

The expression of FCF in year 10 is:

$$FCF_{10} = 35.75 \times (1.02)^9 - 9.75 + 22 = 35.75 \times (1.02)^9 + 12.25$$

The NPV is thus equal to:

$$\begin{aligned} NPV &= -150 + \frac{35.75}{1.12} \sum_{t=0}^8 \left(\frac{1.02}{1.12} \right)^t + \frac{0.25}{1.12} \sum_{t=0}^8 \left(\frac{1}{1.12} \right)^t + \frac{35.75 \times (1.02)^9 + 12.25}{(1.12)^{10}} \\ &= -150 + \frac{35.75}{1.12} \left(\frac{1 - \frac{1.02^9}{1.12^9}}{1 - \frac{1.02}{1.12}} \right) + \frac{0.25}{1.12} \left(\frac{1 - \left(\frac{1}{1.12} \right)^9}{1 - \left(\frac{1}{1.12} \right)} \right) + \frac{(35.75(1.02)^9 + 12.25)}{(1.12)^{10}} \\ &= 72.46 \end{aligned}$$

- d. For discount rates less than or equal to the IRR. The IRR is the discount rate that equalizes NPV to 0. Plot:

$$-150 + 36 \left(\frac{1 - \frac{1}{(1+x)^{10}}}{1 - \frac{1}{1+x}} - 1 \right) + \frac{48}{(1+x)^{10}}$$